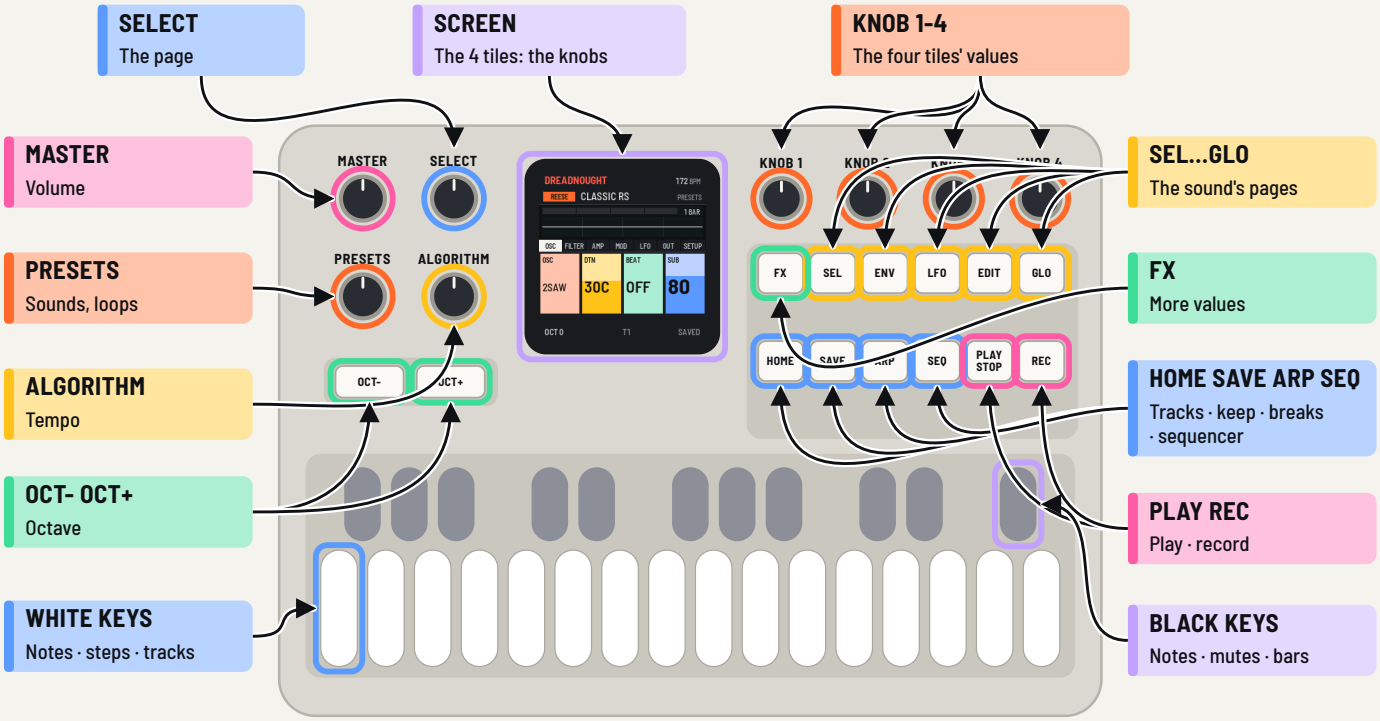
# DREADNOUGHT

## Dark jungle & drum and bass firmware for the M-VAVE FM-1

| FIRMWARE | TRACKS | INSTRUMENTS | PATTERNS |
|---|---|---|---|
| 0.36 | 8 | 8 | 64 |

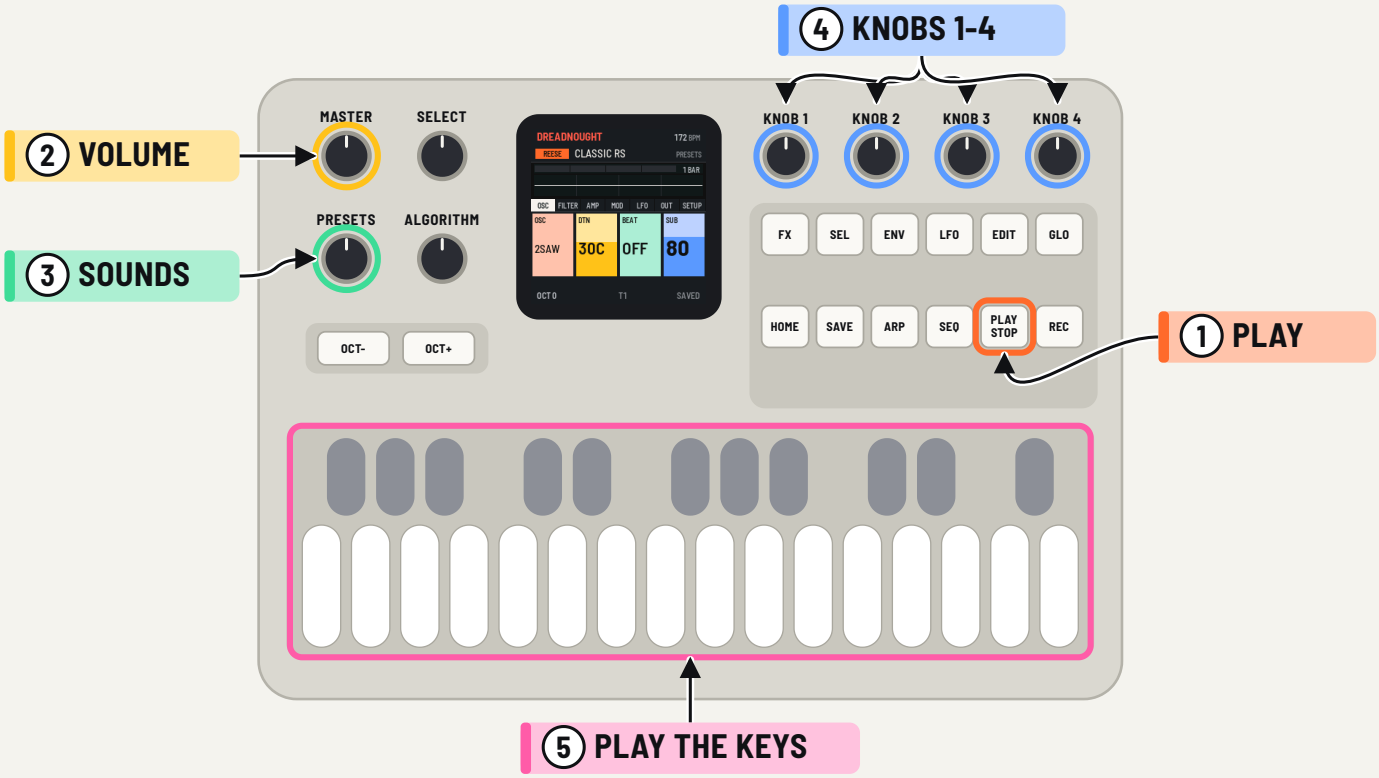



Manual & quick start · github.com/TwistedGenetics/DREADNOUGHT · a beta: back up your FM-1 first (see INSTALLING, UPDATING, RECOVERY)

# QUICK START

Six walks round the box, each a few presses. Every one starts from the screen DREADNOUGHT starts on: REESE's OSC page. The numbers on the pictures are the order to press things in.

## 1 HEAR IT



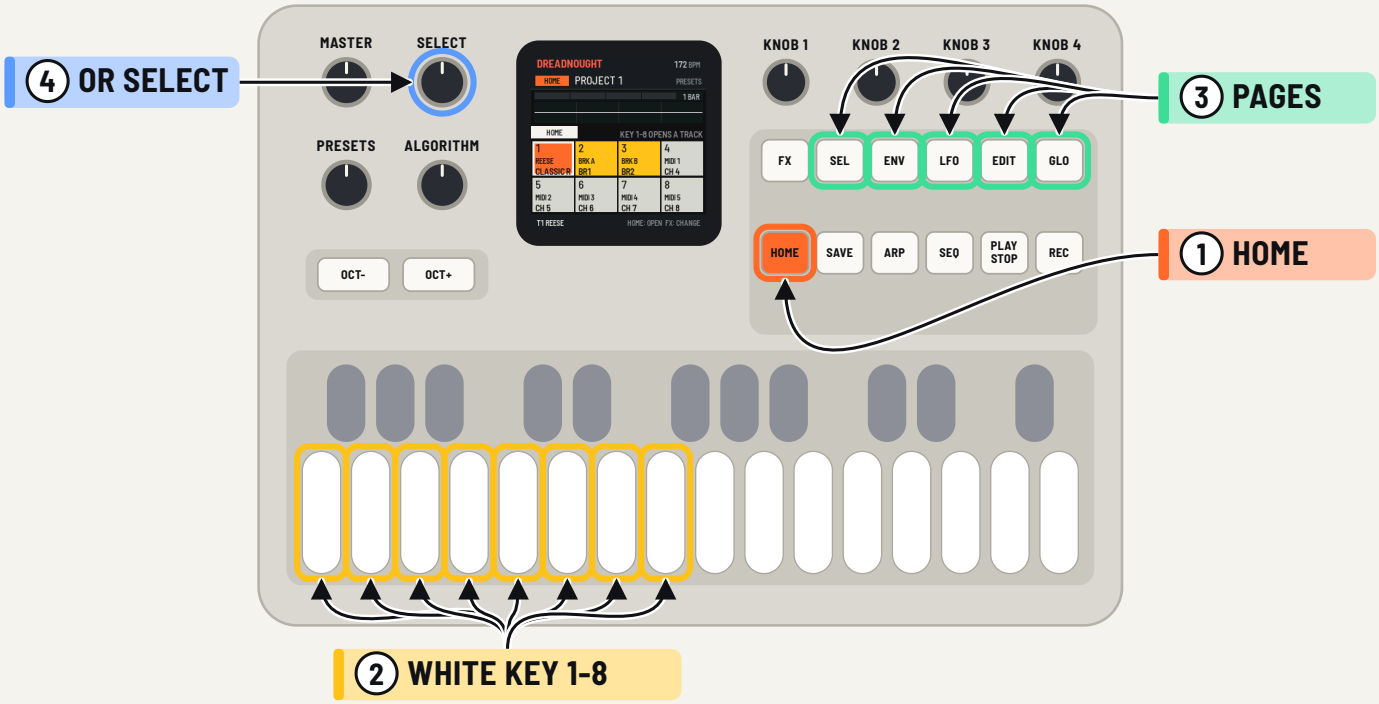

1. Press **PLAY**. The demo pattern plays: REESE, the bass on track 1. **PLAY** again stops it.
2. Turn **MASTER** for the volume.
3. Turn **PRESETS**: REESE's factory sounds, one a detent (CLASSIC RS, OG REESE, HOLLOW RS, TECHSTEP ...). The sound bar names it.
4. Turn **KNOB 1** – **KNOB 4**. Each changes the tile in its colour: orange, yellow, green, blue.
5. Play the keys: notes from F3 up to G5. Hold one and press another: it slides (REESE is mono).

**ON THE SCREEN**




REESE's OSC page: OSC, DTN, BEAT, SUB.

# 2 FIND YOUR WAY




1. Tap **HOME**. The eight tracks, each its number, instrument and sound, in its instrument's colour.
2. Press a white key 1-8 (the first eight from the left): that track's instrument opens on its first page.
3. **SEL** **EDIT** **ENV** **LFO** **GLO** are its pages: OSC, FILTER, AMP, MOD, OUT. Press one again for its second page (REESE: ENV again is the filter envelope).
4. Or turn **SELECT** to go through every page in turn. The tabs show where you are.

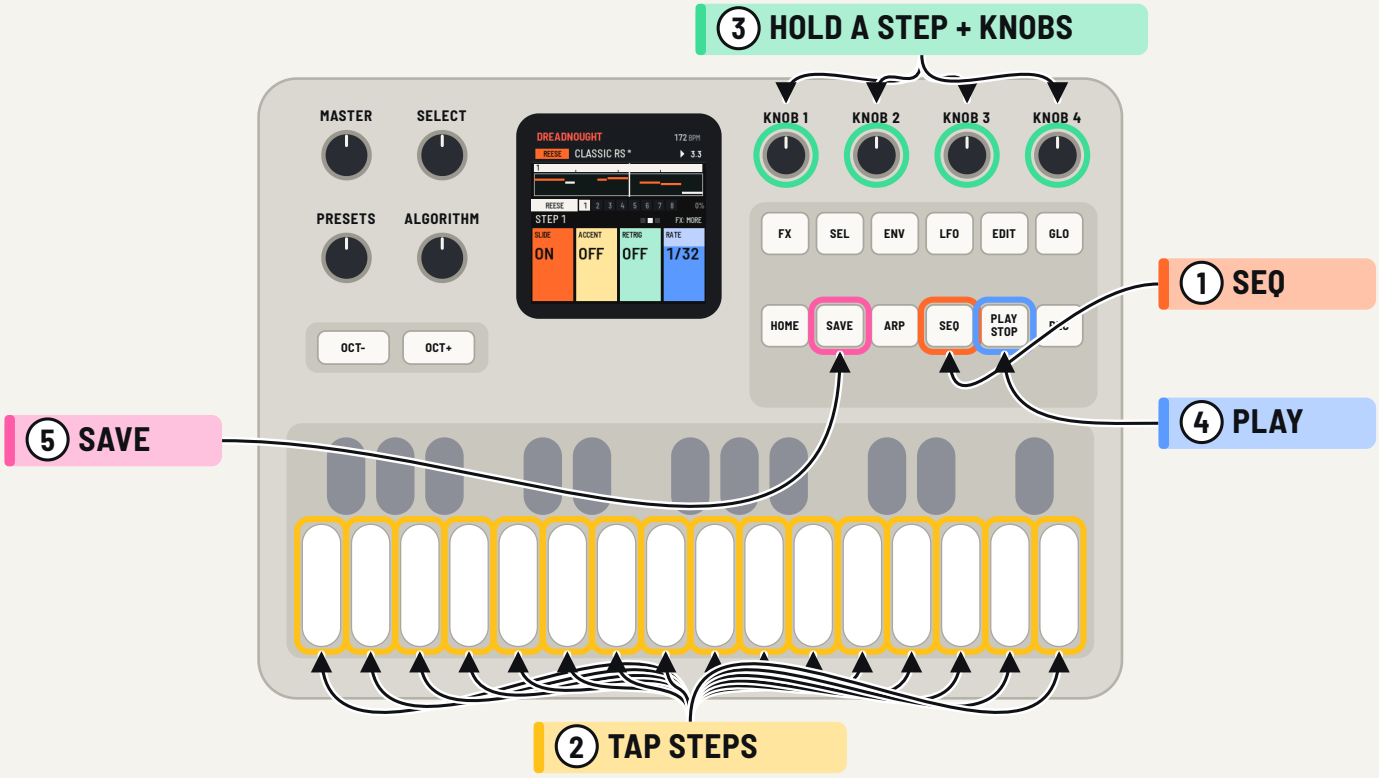
**TIP** From anywhere: hold **SEQ** and press a white key 1-8 to jump to that track's sequencer, or a black key 1-8 to its sound pages.

**ON THE SCREEN**




HOME: the eight tracks.

# 3 MAKE A PATTERN



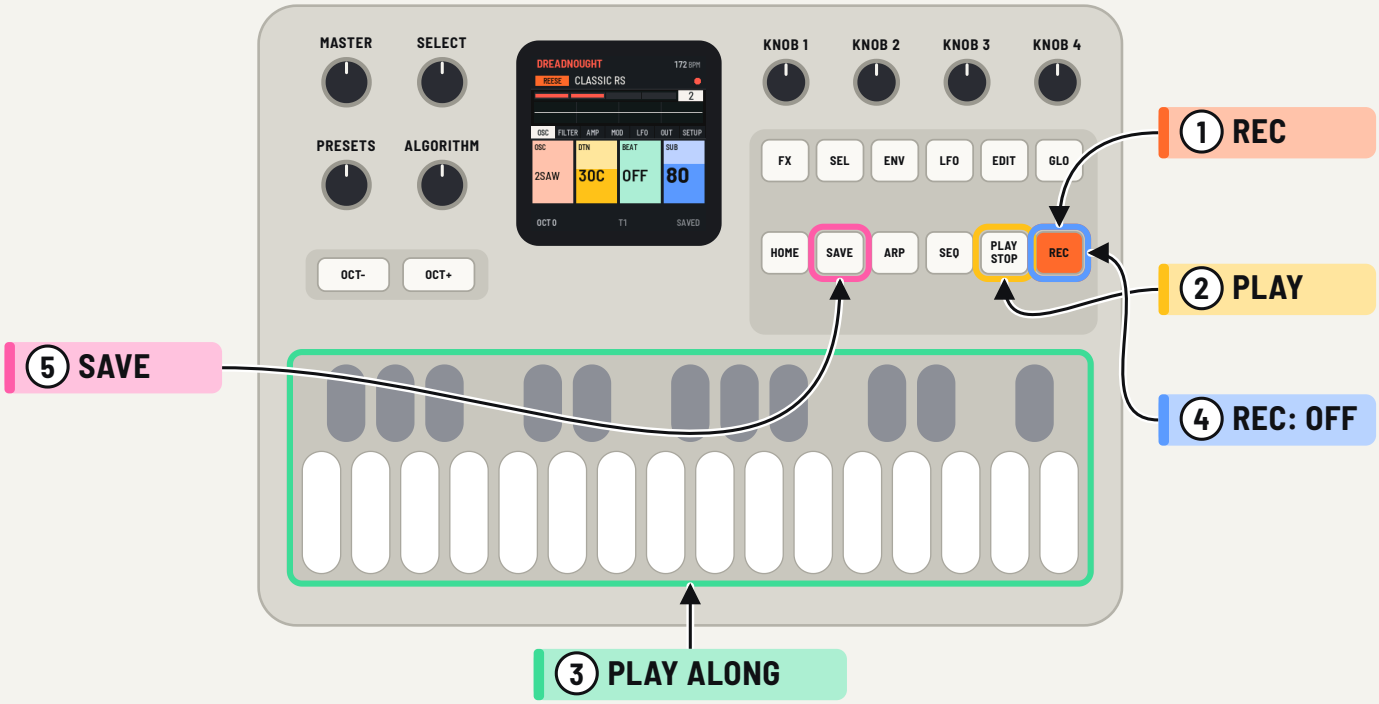

1. On the track you want, press **SEQ**: its piano roll. (On a break: the slice grid, DRUM.)
2. Tap white keys: steps 1-16 on and off. A new step gets the note, length and chord you last put in.
3. Hold a step: its four values pop up big. Turn **KNOB 1** NOTE, **KNOB 2** LENGTH, **KNOB 3** VEL, **KNOB 4** COND (a chance, FILL, every 2nd time…). **FX** while holding: SLIDE, ACCENT, RETRIG, RATE; again: the retrig's fade and CHORD.
4. **PLAY** to hear it. Black keys 10 and 11 show bars 2-4 (64 steps).
5. Tap **SAVE** (lit when something has changed) to keep it.

**ON THE SCREEN**




A step held: its values, big.

# 4 RECORD LIVE




1. Press **REC** (lit).
2. Press **PLAY**: a bar of clicks to count you in, then the pattern runs with a click on the beat.
3. Play: each note lands on its nearest step, as long as you held it. Keys held together on FM, WAVE, SAMPLER's keys or MIDI make a chord step.
4. Press **REC** again to stop recording; **PLAY** to stop.
5. Tap **SAVE**.

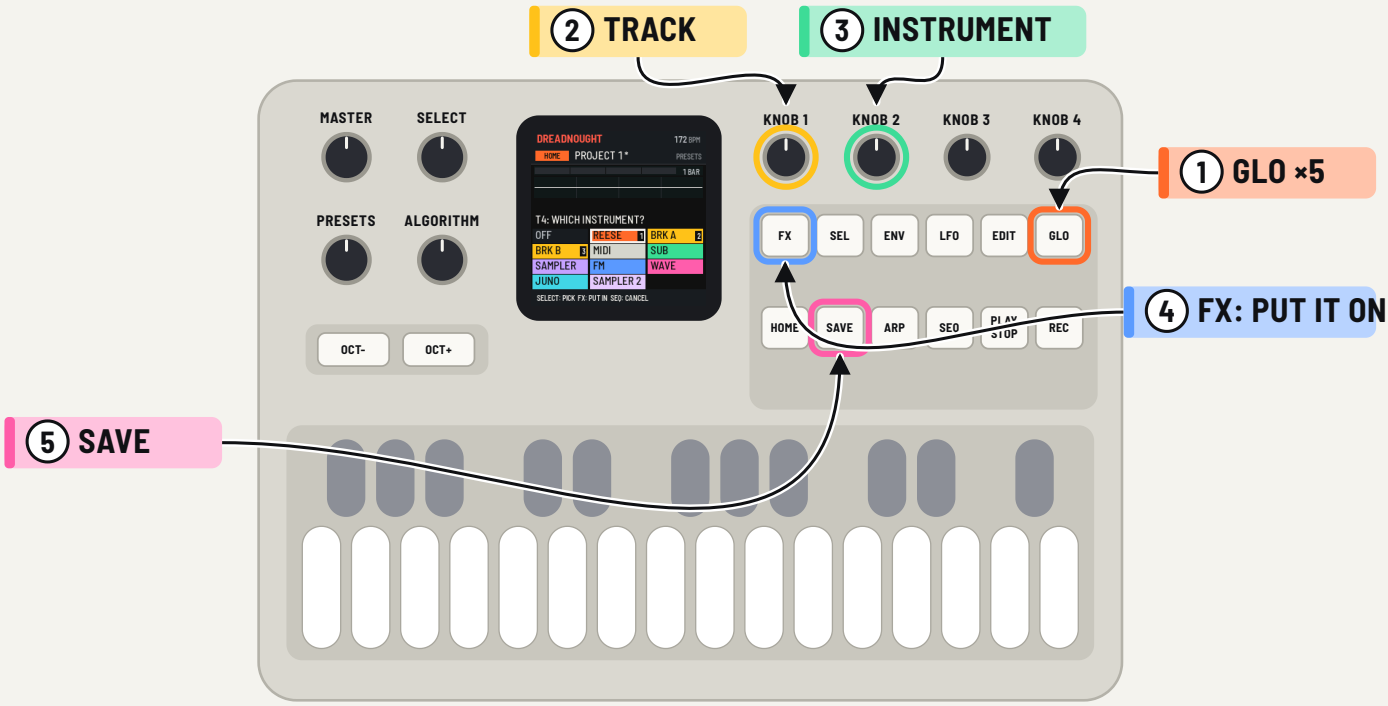
**TIP** Stopped, REC records step by step: each key writes the next step. The click and the count-in are on SETUP (**GLO** three times).

**ON THE SCREEN**



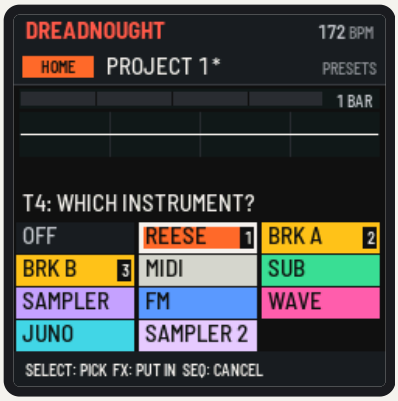

The count-in: the bar counts down.

# 5 CHOOSE THE INSTRUMENTS



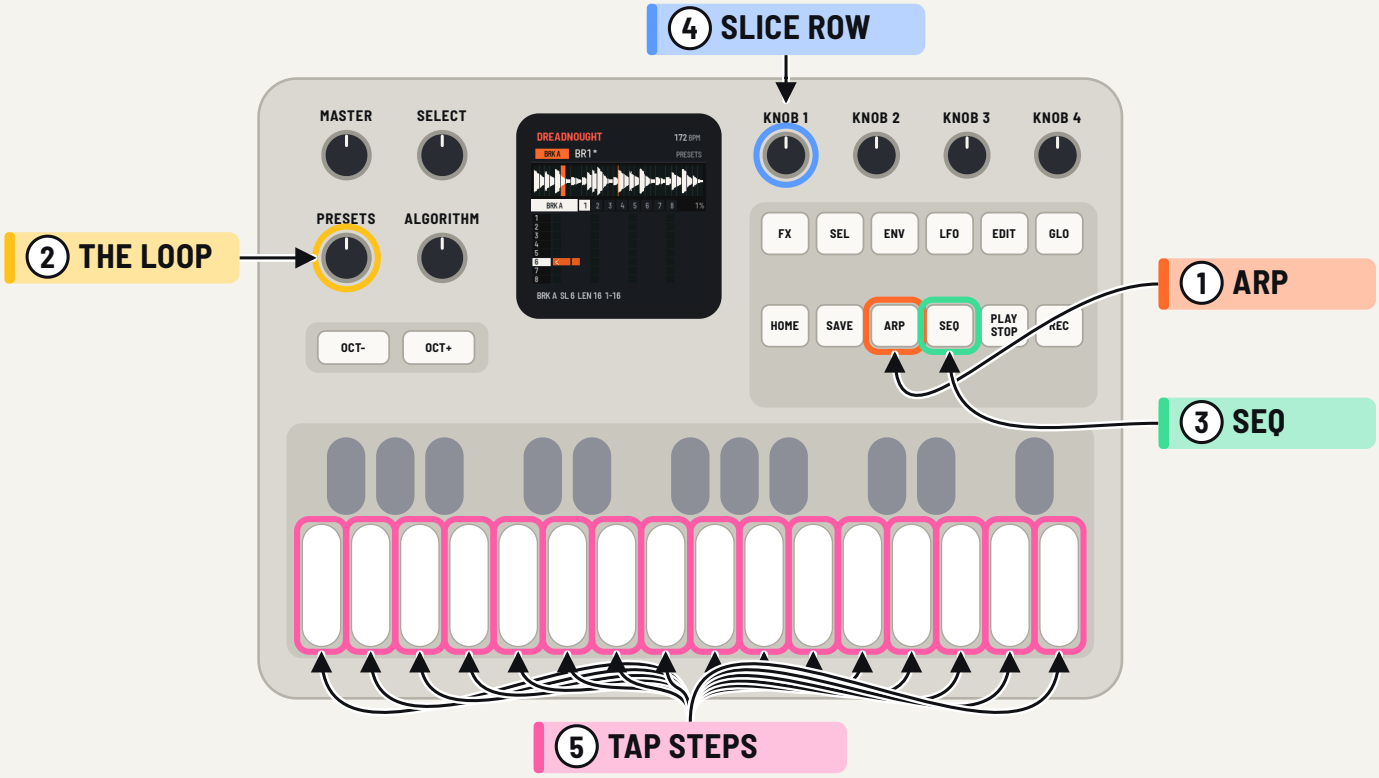

1. Tap **HOME**. Turn **SELECT** to the track.
2. Tap **FX**: the picker. (An empty track, OFF, opens it by itself when you press its key.)
3. Turn **SELECT** or **KNOB 1**: OFF, REESE, BRK A, BRK B, MIDI, SUB, SAMPLER, FM, WAVE, JUNO or SAMPLER 2. One already on another track shows that track's number.
4. **FX** puts it on the track and opens it; **SEQ** calls it off. Each instrument but MIDI is on one track at a time: given to another, the two tracks swap. The patterns stay with their track.
5. **SAVE** keeps the layout.

**ON THE SCREEN**




HOME's picker: track 4, REESE (now on track 1).

# 6 PLAY A BREAK




1. First send some loops: the breaks page or the web editor's Project tab (see THE WEB EDITOR). Until then the breaks are silent.
2. Open a break track (HOME, key 2) and press **ARP**: BRK, the loop's page. **ARP** again: BRK2, then SLICE.
3. Turn **PRESETS** to pick the loop.
4. **SEQ**: the slice grid. **KNOB 1** picks the slice row; tap white keys to put that slice on those steps.
5. **PLAY**. The loop follows the tempo (**ALGORITHM**).

**ON THE SCREEN**




DRUM: slices down, steps across.

# 1 THE PANEL

**Everything on the FM-1, and what it does in DREADNOUGHT. The screen tells you what the four knobs do on each page; the buttons do the same thing everywhere.**

| CONTROL | DOES |
|---|---|
| MASTER | Volume. |
| SELECT | The page (on the sound pages). On the breaks' pages: BREAKS A / B. On MIDI: which MIDI instrument. With a step held: moves it. |
| PRESETS | The track instrument's sounds. On a break: the loop. With SAVE held: the user slot to store into. On ROLL with a step held: its chord. |
| ALGORITHM | The tempo, 60-200 BPM. |
| KNOB 1 – KNOB 4 | The four tiles on the screen, in their colours: orange, yellow, green, blue. |
| OCT- OCT+ | Octave, -3 to +3. With a step held: moves its note an octave. |
| SEL EDIT ENV LFO GLO | The sound's five pages (OSC, FILTER, AMP, MOD, OUT on the synths). Pressed again: the page's second view. GLO goes on: FX, SETUP, MIDI, PROJECT. |
| FX | Tap: more: the next four values of an effect, a held step, an LFO; on HOME, the instrument picker; on SETUP, the skin; on ROLL and DRUM, the keys' mode. Held ½ s: the FX page (again: back). |
| HOME | Tap: the HOME page (the eight tracks). On a sequencer page, held: FILL. |
| SAVE | Tap: keep the sound, patterns, slices and layout (lit when something changed). Held with other controls: store, name, copy (see SAVING). |
| ARP | BREAKS: BRK, again BRK2, again SLICE (chop mode). |
| SEQ | This instrument's sequencer page; again: back. Held + a key 1-8: jump to a track. |
| PLAY | Start / stop the sequencer (MIDI clock, start and stop go out on USB). |
| REC | Record into the selected track: playing, live; stopped, step by step. |
| White keys | Notes. On ROLL and DRUM: the 16 steps in view. On HOME: tracks 1-8. On SLICE with REC lit: chops. |
| Black keys | Notes. On ROLL and DRUM: 1-8 mute tracks 1-8, 9 held 1 s clears the track, 10 / 11 the bar in view. |
| OCT- + OCT+ held 5 s | Update mode, for the flasher (recovery). |

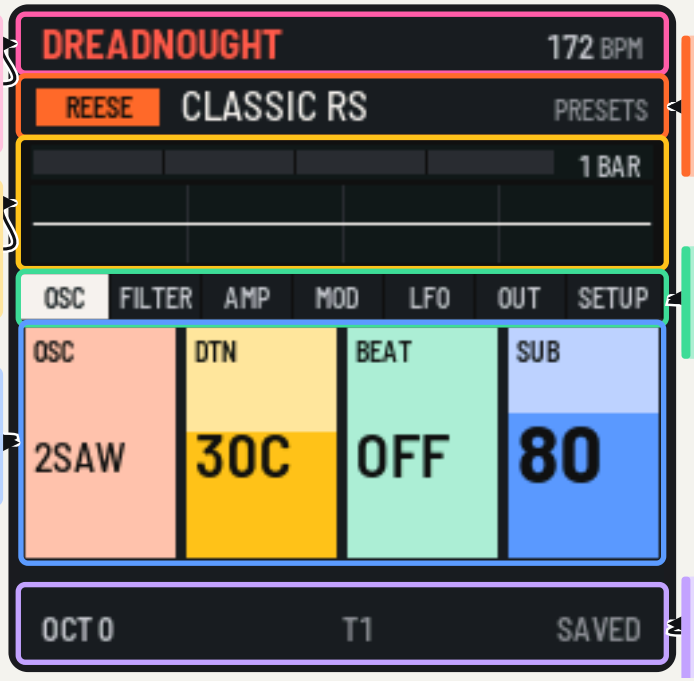
# 2 THE SCREEN

**DREADNOUGHT's screen is built to be read from arm's length: big values, each knob its own coloured tile. This is the default skin, BLOCKS; SETUP has two more (see SETUP AND SKINS).**

**TOP BAR**
DREADNOUGHT, the tempo; playing: the bar and beat.

**BARS AND SCOPE**
The pattern's bars, where it is, the beat; the sound's wave.

**THE FOUR TILES**
KNOB 1-4: name, value, filled as far as the value goes.




**SOUND BAR**
The instrument (REESE) and its sound. * changed since SAVE.

**TABS**
The sound's pages; lit: this one.

**STATUS LINE**
Octave, track, SAVED; messages.

The tile fills from the bottom as far as the value goes; a value that can go either way (a pitch, an envelope amount) fills from the middle. When something happens (a refusal, a pattern queued, "STORE TO USER 3") the status line says so for a moment.




HOME: the tracks in their instrument's colour.




A step held on ROLL: its values big.




PROJECT: memory and CPU.

# 3 TRACKS, HOME AND PROJECTS

**A project is eight tracks. Each plays one instrument and has 8 patterns of up to 64 steps.**

Out of the box: track 1 REESE, tracks 2 and 3 BREAKS A and B, tracks 4-8 MIDI (your other gear, channels 4-8). HOME changes that (quick start 5); the instruments you can put on a track:

| INSTRUMENT | WHAT IT IS |
|---|---|
| **REESE** | Detuned saws through a ladder filter: the jungle bass. Mono, slides. |
| **SUB** | A 101-style mono synth with a dirt stage: dread bass, acid, 808 subs. |
| **BRK A, BRK B** | Two break players: loops in time with the tempo, sliced and re-sequenced, chopped by hand. |
| **SAMPLER** | 16 one-shot pads, or a key map of 8 zones (a multisample). |
| **FM** | A six-operator DX7-style synth (Dexed's engine): plays DX7 voices. 6 voices. |
| **WAVE** | A wavetable synth: 8 tables, your own Serum tables, unison, warps. 6 voices. |
| **JUNO** | A Juno-60 / 106-style polysynth with its chorus. 6 voices. |
| **SAMPLER 2** | A second sampler: its own pads or key map. |
| **MIDI** | Plays other gear on its channel over USB. Any number of tracks. |
| **OFF** | Plays nothing. |

## HOME

Tap HOME from any sound page. Each track shows its number, instrument and sound; a dot lights when it plays a note, a muted track is dim. A white or black key 1-8 opens that track. Or turn SELECT to pick one and tap HOME again. Hold HOME and turn SELECT to step through the instruments without leaving the page you're on.

FX on HOME opens the instrument picker for the selected track: turn SELECT or KNOB 1, FX puts it in and opens it, SEQ calls it off. An empty track (OFF) opens the picker when you open it. PROJECT ( GLO ×5) shows the samples' and effects' memory and the CPU load.

## JUMPING

| | |
|---|---|
| SEQ **held + white key 1-8** | That track's instrument (on a sequencer page: its sequencer page). |
| SEQ **held + black key 1-8** | That track's sound pages, from anywhere. |
| HOME **held +** SELECT | The next / previous instrument's pages. |

## SAVING

SAVE keeps everything: each instrument's sound, all 64 patterns, the layout, the effects, your slices. It comes back at power-on. The project has a name: on PROJECT, hold SAVE and press FX to name it (see SAVING). The web editor saves whole projects to your computer.

# 4 REESE

**Up to seven drifting saws, a Moog-style ladder with drive, a swept comb for the hollow movement and a clean sine on the fundamental. Mono: a key held under another slides.**

| PAGE | GET THERE | KNOB 1 | KNOB 2 | KNOB 3 | KNOB 4 |
|---|---|---|---|---|---|
| **OSC** | SEL | OSC: 2SAW … 7SAW, SAWSQ, 2SIN | DTN: detune | BEAT: the movement locked to the tempo | SUB: the sine |
| **FILTER** | EDIT | CUT | RES | DRV: drive | NTCH: the comb |
| **AMP** | ENV | ATK | DEC | SUS | REL |
| **F.ENV** | ENV again | F.ATK | F.DEC | F.SUS | F.REL |
| **MOD** | LFO | FENV: filter env amount | GLIDE (ms) | L1 RATE | L1 DEPTH |
| **OUT** | GLO | LEVEL | TRANS | BPM | OVERLAP: SLIDE / CUT |

PRESETS: CLASSIC RS, OG REESE, HOLLOW RS, TECHSTEP, NEURO RS, SYNC RS, DARK RS, SUB REESE, HOOVER RS, INIT, then USER 1-8. FX on AMP: ENV SYNC, the envelope's times as note lengths at the tempo.




OSC.




FILTER.




F.ENV (ENV again).

**TIP** OVERLAP: SLIDE glides a note into the next one when its length reaches past it (no new attack): the classic rolling reese. CUT retriggers every note.

# 5 SUB

**A 101-style mono synth: saw, pulse with PWM, a sub-oscillator and noise into a resonant filter that tracks the keys, then the dirt.**

| PAGE | GET THERE | KNOB 1 | KNOB 2 | KNOB 3 | KNOB 4 |
|---|---|---|---|---|---|
| **OSC** | SEL | SAW | PULSE | SUB | NOISE |
| **OSC2** | SEL again | PW | PWM | PWM RATE | SUB MODE: -1 OCT / -2 OCT / -2 25% |
| **FILTER** | EDIT | CUT | RES (self-oscillates) | ENV | KEY: tracking |
| **DIRT** | EDIT again | DRIVE | TYPE: SOFT / HARD / FOLD / RECT | CRUSH | TONE |
| **AMP** | ENV | ATK | DEC | SUS | REL |
| **F.ENV** | ENV again | F.ATK | F.DEC | F.SUS | F.REL |
| **MOD** | LFO | GLIDE | PORTA: OFF / AUTO / ON | ACCENT | VCA: ENV / GATE |
| **OUT** | GLO | LEVEL | TRANS | SPLIT: below it, no dirt | OVERLAP |

PRESETS: DREAD, DONK, ROLLER, WOBBLE, ACID 101, 808 SUB, GROWL, SQUARE SB, PUNCH, TECH SUB, CRUSHED, RAVE STAB, INIT, then USER 1-8. ACCENT plays the sequencer's ACCENT steps louder and brighter, as the 303 does.




OSC.




DIRT (EDIT again).




MOD.

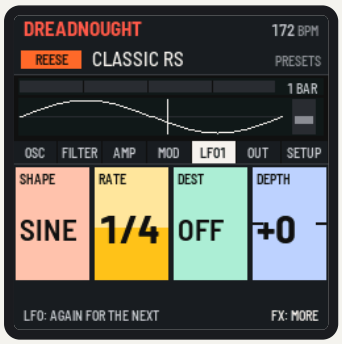
# 6 LFOS

**REESE and SUB each have four LFOs, part of their sound: a preset brings its own.**

Press LFO on REESE or SUB: MOD. Again: LFO 1, then 2, 3, 4, then back to MOD. FX shows the LFO's MORE values, and on a DRAW LFO, the drawing.

| PAGE | GET THERE | KNOB 1 | KNOB 2 | KNOB 3 | KNOB 4 |
|---|---|---|---|---|---|
| **LFO 1-4** | LFO again | SHAPE: SINE TRI SAW UP SAW DN SQUARE S&H SMOOTH DRAW | RATE: Hz, or a note length | DEST: where it goes | DEPTH: + or - |
| **MORE** | FX | SYNC: FREE / SYNC | MODE: FREE / RETRIG / 1-SHOT | PHASE | SMOOTH |
| **DRAW** | FX again | the pen (a point's value) | SMOOTH | POINTS: 16 / 32 | points in view |

DEST: the synth's own values (CUTOFF, RES, DRIVE, NOTCH, DETUNE, PITCH, SUB, LEVEL, F.ENV; SUB's S CUT, S RES ...), the breaks' (A PITCH, A DECAY ...) or another LFO's depth. DRAW: the white keys are the points; press one to set it to the pen.




An LFO.




MORE (FX).




DRAW: a shape by hand.

# 7 BREAKS

**Two break players, A and B, on tracks 2 and 3 unless you move them. They play the loops you send from the editor or the breaks page, always in time.**

| PAGE | GET THERE | KNOB 1 | KNOB 2 | KNOB 3 | KNOB 4 |
|---|---|---|---|---|---|
| **BRK** | ARP | LOOP (PRESETS too) | PITCH | DECAY (127: hold) | CRUSH |
| **BRK2** | ARP again | CHOKE: group OFF, 1-4 | SLICES: GRID / AUTO / MANUAL | STRETCH: REPITCH / SLICE / STRETCH | LEVEL |
| **SLICE** | ARP again | CHOP: select | CURSOR (coarse) | FINE | ZOOM |

SELECT on these pages: BREAKS A or B. A and B choke each other (group 1) like one sampler voice.



BRK.



DRUM, playing.



SLICE: chops.

## THE SLICE GRID (DRUM)

SEQ from a break's page. Each row is a slice, each column a step. KNOB 1 picks the row; tap white keys to put it on steps. KNOB 2 the track's length, KNOB 3 swing, KNOB 4 the steps in view. Hold a step: SLIDE plays it reversed. SAVE + FX: the break laid out straight, each step its own slice.

## CHOP MODE (SLICE)

Set SLICES to MANUAL on BRK2, then ARP again. The MPC way: press PLAY, press REC (lit: the keys cut) and hit the lowest key on the first hit, the next key on the next, and so on. Each chop lands where you hit it; hit a key again to retake it. REC off: the keys play the chops. Chops save themselves.

| ON SLICE | DOES |
|---|---|
| KNOB 1 | Select a chop (the cursor jumps there, you hear it). |
| KNOB 2 / KNOB 3 | The cursor: coarse / fine. |
| KNOB 4 | Zoom: the whole loop ... 1/64 of it. |
| FX | Move the selected chop to the cursor. |

| ON SLICE | DOES |
| --- | --- |
| HOME | Delete the selected chop; held 1 s: delete them all. |
| SAVE + REC | A chop between the others, at the cursor. |
| SAVE + OCT- / OCT+ | Start again from AUTO's hits / from the GRID. |

# 8 SAMPLER

**Your samples (sent with the editor) on 16 pads, or mapped across the keys.**

| PAGE | GET THERE | KNOB 1 | KNOB 2 | KNOB 3 | KNOB 4 |
|---|---|---|---|---|---|
| **PAD** | SEL | SAMPLE | START | END | REVERSE |
| **FILTER** | EDIT | CUT | RES | DRIVE | CRUSH |
| **AMP** | ENV | ATTACK | HOLD (127: to the end) | DECAY | LEVEL |
| **MOD** | LFO | PITCH | FINE | PAN | CHOKE |
| **OUT** | GLO | MODE: ONE-SHOT / KEYS | PAD | BPM | |

ONE-SHOT: 16 pads, played by the first 16 keys and sequenced on DRUM's rows. PRESETS: the pad shown.

| PAGE | GET THERE | KNOB 1 | KNOB 2 | KNOB 3 | KNOB 4 |
|---|---|---|---|---|---|
| **ZONE** | SEL | SAMPLE | ROOT | LOW | HIGH |
| **FILTER** | EDIT | CUT | RES | DRIVE | CRUSH |
| **AMP** | ENV | ATTACK | DECAY | SUSTAIN | RELEASE |
| **LOOP** | LFO | LOOP: OFF / LOOP / SUST | L.START | L.END | TUNE |
| **OUT** | GLO | MODE | ZONE | LEVEL | VOICES (1: mono) |

KEYS: 8 zones, each a sample over a range of keys. Sequenced on ROLL; plays chords.




ONE-SHOT: a pad.




The pads on DRUM.




KEYS: a zone.

# 9 FM

**A six-operator FM synth that plays DX7 voices: ten factory voices, and any you keep from a DX7 bank in the editor.**

| PAGE | GET THERE | KNOB 1 | KNOB 2 | KNOB 3 | KNOB 4 |
|---|---|---|---|---|---|
| **VOICE** | SEL | ALGO 1-32 | FEEDBACK | BRIGHT: the modulators | TIME: every envelope |
| **OP** | EDIT | OP: which of the six | COARSE (ratio / Hz) | FINE | LEVEL |
| **OP+** | EDIT again | OP | DETUNE | VEL | MODE: RATIO / FIXED |
| **EG** | ENV | R1 | R2 | R3 | R4 |
| **EG LVL** | ENV again | L1 | L2 | L3 | L4 |
| **LFO** | LFO | SPEED | DELAY | PMD | AMD |
| **LFO+** | LFO again | WAVE | SYNC | PMS | KEY SYNC |
| **OUT** | GLO | LEVEL | VOICES 1-6 (1: mono) | ATTACK | RELEASE |

PRESETS: DUB FM BS, FM REESE, HOLLOW STB, DARK BELL, METAL HIT, FM PLUCK, DARK PAD, GROWL BASS, SINE SUB, INIT VOICE, then your DX7 voices. BRIGHT, TIME, ATTACK and RELEASE sit on top of the voice without changing it.




VOICE.




OP: the operator's frequency.




EG: OP3's rates.

# 10 WAVE

**A wavetable synth with two oscillators (0.33): 8 factory tables and your own Serum tables, A and B, FM, ring, AM and sync between them, warps, six filters, seven drives, three envelopes, three LFOs and a matrix of 8 routes.**

Each page button pressed again shows the page's next view; the tab names it. SEL: WAVE A, OSC A, WAVE B, OSC B, B MIX. EDIT: FILTER, DRIVE. ENV: AMP, MOD ENV, ENV 3. LFO: MOD, LFO 1, LFO 2, LFO 3, MATRIX.

| PAGE | GET THERE | KNOB 1 | KNOB 2 | KNOB 3 | KNOB 4 |
|---|---|---|---|---|---|
| **WAVE A** | SEL | TABLE | POS: through the frames | WARP: OFF SYNC BEND MIRROR PWM FM CRUSH | AMOUNT |
| **OSC A** | SEL ×2 | UNISON 1-4 | DETUNE | SPREAD: stereo | SUB: a sine an octave down |
| **WAVE B** | SEL ×3 | B TABLE | B POS | B WARP | B AMT |
| **OSC B** | SEL ×4 | OSC B: OFF / ON | B OCT | B SEMI | B FINE |
| **B MIX** | SEL ×5 | B LEVEL: B heard | B MODE: MIX FM RING AM SYNC | MOD AMT | NOISE |
| **FILTER** | EDIT | FILTER: LP12 LP24 HP BP NOTCH COMB | CUT | RES | ENV: mod env on the cutoff |
| **DRIVE** | EDIT ×2 | DIST: SOFT HARD FOLD RECT TUBE BITS RATE | DRIVE | SAT: after the filter | KEY: cutoff follows the note |
| **AMP** | ENV | ATK | DEC | SUS | REL |
| **MOD ENV** | ENV ×2 | M.ATK | M.DEC | M.SUS | M.REL |
| **ENV 3** | ENV ×3 | E3 ATK | E3 DEC | E3 SUS | E3 REL |
| **MOD** | LFO | POS ENV | POS LFO | L1 RATE | CUT LFO |
| **LFO 1-3** | LFO ×2-4 | SHAPE: SINE TRI SAW SQUARE S&H RANDOM | RATE: Hz, or a length | SYNC: FREE / SYNC | MODE: FREE / RETRIG |
| **MATRIX** | LFO ×5 | SLOT 1-8 | SOURCE | DEST | DEPTH |
| **OUT** | GLO | LEVEL | VOICES 1-6 (1: MONO) | GLIDE | TRANS |

PRESETS: INIT, SUPERSAW, REESE, PPG BELL, VOX PAD, SYNC LEAD, PWM STRNG, GRIT BASS, ORGAN, FM GROWL, BENT KEYS, CRUSHER, FM BELL, SYNC SWEEP, RING DRONE, COMB PLUCK, FOLD BASS, HOOVER, BIT WOBBLE, S&H

TWIST.



WAVE A.



B MIX (SEL ×5).



DRIVE (EDIT again).



MATRIX (LFO ×5).

| B MODE | WHAT IT DOES |
|---|---|
| MIX | B is only heard, at B LEVEL. |
| FM | B moves A's phase: MOD AMT is how far. Bells, growls, metal. B LEVEL 0 to hear only A. |
| RING | A times B: their sum and difference, A's own pitch gone at MOD AMT 127. Clangs and drones. |
| AM | B moves A's level: tremolo at low B pitches, sidebands at high ones. |
| SYNC | A starts again with each of B's cycles. A plays MOD AMT above B (up to 4 octaves): sweep MOD AMT (the matrix: MOD ENV to MOD AMT) for the classic sync scream. |

| MATRIX | |
|---|---|
| Sources | LFO 1-3, MOD ENV, ENV 3, AMP ENV, VEL (velocity), KEY (the note), RANDOM (one per note). |
| Destinations | POS A, POS B, WARP A, WARP B (their AMOUNTs), PITCH, PITCH B, MOD AMT, B LEVEL, CUTOFF, RES, DRIVE, LEVEL, PAN, SUB, NOISE, L1 RATE, L2 RATE, DETUNE. |
| Depth | -64 to +63: how far, either way. Routes add up. |

| TABLE | POS 0 → 127 |
|---|---|
| ANALOG | Sine, to triangle, to square, to saw. |
| REESE | A saw with notches that move: two detuned saws, frozen. |
| PPG | A band of harmonics stepping up, on a thin saw. |
| VOCAL | The vowels A E I O U, one into the next. |
| SYNC | Hard sync: a saw restarting 1 to 8 times a cycle. |
| PWM | A pulse from 50 % down to 4 %. |
| GRIT | A sine folded harder and harder. |

| TABLE | POS 0 → 127 |
| --- | --- |
| HOLLOW | Odd harmonics, drawbars moving: an organ going hollow. |

**VOICES** The voices WAVE plays depend on what's switched on, so the FM-1 never runs out of time: UNISON, OSC B, a DIST, LP24, SAT and stereo (SPREAD, PAN) each cost some. Plain: 6 voices. UNISON 2: 3, UNISON 4: 2. OSC B on: 3, and unison 2 at most. B on with a drive: 2. VOICES is the most it will use; watch CPU on PROJECT.

# 11 JUNO

**A Juno-60 / 106-style polysynth (0.36): one DCO a voice with saw, pulse, sub and noise, the HPF, a resonant 4-pole filter, one envelope, one LFO with delay, and the chorus. Six voices.**

SEL, EDIT and ENV pressed again show the page's second view; the tab names it.

| PAGE | GET THERE | KNOB 1 | KNOB 2 | KNOB 3 | KNOB 4 |
|---|---|---|---|---|---|
| **DCO** | SEL | RANGE: 16' 8' 4' | SAW: OFF / ON | PULSE: OFF / ON | SUB: an octave down |
| **DCO 2** | SEL ×2 | PWM: how far | PWM SRC: LFO MAN ENV | NOISE | LFO: on the pitch |
| **VCF** | EDIT | CUT | RES (rings on its own near the top) | ENV | KEY: follows the keys |
| **VCF 2** | EDIT ×2 | HPF: BOOST FLAT 2 3 | ENV POL: NORM / INV | LFO: on the cutoff | VEL: on the cutoff |
| **ENV** | ENV | ATK | DEC | SUS | REL |
| **VCA** | ENV ×2 | VCA: ENV / GATE | VEL: on the level | | |
| **LFO** | LFO | RATE | DELAY: fades in after the first key | SYNC: OFF or a length | MODE: FREE / KEY |
| **OUT** | GLO | CHORUS: OFF I II I+II | LEVEL | VOICES 1-6 (1: mono) | TRANS |

PRESETS: INIT, JUNO PAD, BRASS, STRINGS, 106 BASS, ORGAN, RAVE STAB, PLUCK, PIANO, SUB BASS, ACID POLY, GLASS, DARK PAD, NOISE SWEEP, WHISTLE, HOOVERISH, JUNGLE STAB.




DCO.




VCF.




OUT: the chorus.

**CHORUS** The Juno's sound is the chorus: I is slow and wide, II faster, I+II a quick shimmer. Stereo only with it on. For the 106 bass: PULSE with PWM by hand (MAN), SUB up, RANGE 16', VOICES 1.

# 12 SAMPLER 2

**A second sampler (0.36), on a track of its own: its own 16 pads or key map, its own settings. Put it on a track from HOME's picker. Its pages are SAMPLER's (see SAMPLER); the top line says SMPL 2.**




SAMPLER 2 in KEYS, beside SAMPLER's pads.




HOME: SAMPLER on 6, SAMPLER 2 on 7.

# 13 MIDI TRACKS

**Tracks set to MIDI play your other gear over USB, each on its channel.**

| PAGE | GET THERE | KNOB 1 | KNOB 2 | KNOB 3 | KNOB 4 |
|---|---|---|---|---|---|
| **MIDI** | GLO ×4 | MIDI: which (1-5…) | CHANNEL 1-16 | OVERLAP: SLIDE / CUT | |

SELECT on MIDI: the next MIDI instrument. With a MIDI track selected the keys play out on its channel. SLIDE sends the new note before the old one's note-off: legato on the other synth.

# 14 THE SEQUENCER

**Every instrument has its own sequencer page: ROLL (a piano roll) for the synths, SAMPLER's keys and MIDI; DRUM (a slice grid) for the breaks and SAMPLER's pads.**




ROLL: REESE's pattern.




Marks: conditions, retrigs, accents.




DRUM: a break.

| ON ROLL / DRUM | DOES |
|---|---|
| White keys | The 16 steps in view: tap on / off. |
| Black keys 1-8 | Mute / unmute tracks 1-8 (lit: playing). |
| Black key 9, held 1 s | Clear the track. |
| Black keys 10 / 11 | The bar in view: steps 1-16, 17-32, 33-48, 49-64. |
| Black key 10 or 11 held + white key 1-4 | The track 1, 2, 3 or 4 bars long. |
| KNOB 1 – KNOB 4 (ROLL) | Scroll the pitches, transpose the track, its length (1-64), the steps in view. |
| KNOB 1 – KNOB 4 (DRUM) | The slice row, length, swing, the steps in view. |
| FX (ROLL) | The keys' mode, round: STEP (keys are steps), KEYS (notes at a cursor: KNOB 2 their length, KNOB 4 the cursor, HOME a rest), PLAY (the keys play the sound; REC lit: they record). |
| FX (DRUM) | PLAY mode on / off: the keys play the slices or pads; REC lit: they record. |
| REC held + any knob | REC's grid: live notes to the nearest 1/16, 1/8, 1/4, 1/2 or bar. |
| HOME held | FILL, for FILL / !FILL steps. |

## A STEP HELD

Hold a step's key: its values pop up big, four tiles, one a knob. FX (still holding) shows the next four.

| PAGE | GET THERE | KNOB 1 | KNOB 2 | KNOB 3 | KNOB 4 |
|---|---|---|---|---|---|
| 1 | hold a step | NOTE (break: slice) | LENGTH | VEL | COND |
| 2 | FX | SLIDE (break: reverse) | ACCENT | RETRIG | RATE |
| 3 | FX again | RT LEN | RT VEL: fade in / out | CHORD | |

| DOING | HOW |
|---|---|
| **Move a note** | Hold its step and press the step it should go to (black keys 10 / 11: another bar; onto a note: they swap). Or turn SELECT. OCT- / OCT+: an octave. |
| **Copy a note** | As moving, with SAVE held. |
| **Velocity ramp** | Hold the first step, hold REC, press the last; KNOB 3 / 4 the ramp's start and end. |
| **Retrigs** | RETRIG on: the step repeats at RATE (1/16 a step, 1/32 two …) for RT LEN, its velocity fading (RT VEL): rolls and builds. |
| **Chords** | FM, WAVE, SAMPLER's keys and MIDI: 15 chords (MIN, MAJ, MIN7, MAJ7, 7, MIN9, MAJ9, ADD9, SUS2, SUS4, 5, DIM, AUG, MIN6, 6). Hold a step and turn PRESETS, or CHORD on page 3, or record keys held together. |
| **Conditions** | COND: a % chance; FILL / !FILL; PRE / !PRE (the track's last condition); NEI / !NEI (the track before's); 1ST / !1ST; LST / !LST; A:B (the A-th of every B passes). |
| **Overlaps** | A note whose length reaches the next slides into it (OVERLAP SLIDE on OUT) or retriggers it (CUT). |

## PATTERNS

Each instrument has 8 patterns, along the top of its sequencer page: the playing one lit, a queued one flashing, ones with notes marked, and how full pattern memory is.

| COMBINATION | DOES |
|---|---|
| SAVE + white key 1-8 | This instrument's pattern 1-8: stopped, at once; playing, when the current one ends. |
| SAVE + black key 10 / 11 | Copy this pattern / paste into this one. |
| SAVE + OCT- / OCT+ | Copy / paste the bar in view. |
| SAVE + REC | ×2: the bars with notes copied into the empty bars after them; a full track doubles (16 → 32 → 64 steps). |

## RECORDING

REC lit, playing: each key writes a note at the nearest step, as long as you hold it. Stopped: step by step from the cursor; a key held under another writes a slide. Keys held together make a chord step on the instruments that play chords. A new step you tap in takes the last note, length and chord you used on that track.

To play in while you watch the pattern, put ROLL or DRUM in PLAY mode (FX): the keys play the sound, and with REC lit they record; the bar in view follows the one playing. Hold REC and turn any knob for REC's grid (record quantise): 1/16 (a step), 1/8, 1/4, 1/2 or BAR; the status line shows it as Q 1/4. ROLL's top has a ruler of the pattern's bars (the longest track's), the bar playing lit, so you can see which bar of the whole you're putting notes into.




ROLL in PLAY mode: Q, REC's grid, on the status line.




A 1-bar track against a 4-bar break: the ruler, bar 3 playing.




DRUM in PLAY mode: the key played selects its row.

# 15 EFFECTS

**Two effect slots, FX A and FX B, fed by each track's sends. A master limiter keeps the mix from clipping.**

| PAGE | GET THERE | KNOB 1 | KNOB 2 | KNOB 3 | KNOB 4 |
|---|---|---|---|---|---|
| **MIX** | GLO ×2 | SEND A (the track) | SEND B | RETURN A | RETURN B |
| **FX A** | FX (on the FX page; hold FX anywhere to get there) | the effect | its 1st value | 2nd | 3rd |
| **FX A 2** | FX again | its 4th | 5th | 6th | 7th |
| **FX B, FX B 2** | FX again | as FX A | | | |

176 KB of effect memory: an effect that won't fit beside the other is refused. DELAY fits beside anything; two of the big reverbs fit; two DELAYs don't.




MIX: the sends and returns.




FX A: MASSIVE.




QUADRA, FLANGE.

| EFFECT | WHAT IT IS | VALUES |
|---|---|---|
| DELAY | The dark half-rate ping-pong. | TIME (synced), FEEDBACK, TONE |
| TAPE | Tape echo: wow, flutter, head loss, saturation. | TIME, FEEDBACK, DEGRADE, PINGPONG |
| BBD | Bucket brigade: crumbly, dark. | as TAPE |
| SPRING | Two springs: the dub drip. | TIME, DAMP, PREDELAY, LOW CUT |
| TOMB | A small stone room. | as SPRING |
| MOD | A big moving hall. | as SPRING |

| EFFECT | WHAT IT IS | VALUES |
|---|---|---|
| **CELESTIAL** | MOD with an octave-up shimmer. | as SPRING |
| **DOLMEN** | A small dense chamber, bright. | as SPRING |
| **TUMULUS** | A cavern: long and dark. | as SPRING |
| **QUADRA** | Quadraverb-style chains: CHORUS, REVERB, 5EQ>REV, FLANGE, DETUNE, DLY>REV. | CONFIG, DECAY, TIME, FEEDBACK, MOD, TONE, 16-BIT |
| **MASSIVE** | Super-Massive: ROOM, HALL, HALO, SWELL, ABYSS, ECHOVRB, ORBIT, PRISM, PONG, DUB. | MODE, TIME, FEEDBACK, WARP, DENSITY, MOD, TONE |

# 16 SETUP AND SKINS

GLO three times.

| PAGE | GET THERE | KNOB 1 | KNOB 2 | KNOB 3 | KNOB 4 |
|---|---|---|---|---|---|
| SETUP | GLO ×3 | CLICK: OFF / REC / ON | CLK LVL | COUNT-IN: off / 1 / 2 bars | VEL SENS (REESE) |

FX on SETUP changes the skin: BLOCKS (the default: big coloured tiles), LIGHT (warm white, ink, orange), DARK (the first look). SAVE keeps it.



BLOCKS.



LIGHT.



DARK.

# 17 SAVING, USER SOUNDS, NAMES

| DOING | HOW |
|---|---|
| SAVE (tap) | Keep everything for next time. Lit when something changed. |
| SAVE held + PRESETS | Store REESE's or SUB's sound (with its LFOs) in USER 1-8: turn to pick, let go of SAVE to store. The name editor opens. |
| SAVE held + FX | Name it: on PROJECT the project, on a synth the user sound loaded. |
| The name editor | KNOB 1 or SELECT the place, KNOB 2 or PRESETS the letter; SAVE keeps it, HOME leaves it as it was. |

**TIP** FM and WAVE keep their sound with SAVE too; FM's extra voices and WAVE's extra tables live in the sample slots, sent from the editor.

# 18 THE WEB EDITOR AND THE BREAKS PAGE

**In Chrome or Edge, with the FM-1 on USB: twistedgenetics.github.io/DREADNOUGHT/editor**

The editor opens on a big copy of the FM-1: its screen, knobs, buttons and keys. Everything there does what it does on the FM-1, and turning a knob on the FM-1 moves the page. The tabs after it edit in depth:

| TAB | WHAT'S THERE |
|---|---|
| **Pattern** | Every track in a mouse-driven piano roll or slice grid; velocity, conditions, retrigs, chords, the 8 patterns. |
| **REESE, SUB, LFOs** | Every value; LFO shapes drawn with the mouse. |
| **SAMPLER** | The 16 pads or the key map; SAMPLER 2 picked at the top when it's on a track. |
| **FM** | Every value of the voice. DX7 banks: load a .syx, hear each voice here or on the FM-1, keep some, send them. |
| **WAVE** | Every value. Wavetables: load a Serum wavetable or single-cycle WAV, see its frames, hear it, send it: it comes up on TABLE. |
| **JUNO** | Every value, its factory sounds, keys to hear it on the FM-1. |
| **Breaks** | The breaks' settings and slices. |
| **Project** | Each track's instrument and sends, the FX, memory and CPU; samples (WAV, MP3, AIFF ... converted and sent); save and load whole projects. |

**Sample memory:** three slots, 80 + 80 + 64 KB. Samples, loops, DX7 voices and wavetables share it. A wavetable is about 2 KB a frame. The editor sends only what changed.

**The breaks page** (twistedgenetics.github.io/DREADNOUGHT/breaks) sends loops straight to the slots and sets their slices.

> **TIP** If the editor won't connect: close everything else using the FM-1's MIDI port (a DAW, the flasher or the breaks page in another tab), reload, Connect again. Diagnostics at the bottom of the page lists the ports.

# 19 MEMORY AND CPU

| | |
|---|---|
| **Patterns** | 8 per instrument, 64 steps each; notes of all 64 patterns share memory (about 1 680). The sequencer page shows how full it is. |
| **Samples** | 224 KB in three slots, shared by loops, samples, DX7 voices and wavetables. |
| **Effects** | 176 KB between FX A and FX B. |

| | |
|---|---|
| CPU | PROJECT shows the load and its peak (CPU 40 PK 62%). Too much at once: the status line says AUDIO LATE: CPU FULL. Fewer voices, less unison or a smaller reverb bring it back. |

# 20 INSTALLING, UPDATING, RECOVERY

1. Open twistedgenetics.github.io/DREADNOUGHT/flasher in Chrome or Edge, the FM-1 on USB.
2. Back up first: the flasher saves your FM-1's own firmware.
3. Pick the newest release (or a .fwsc file from GitHub Releases) and press Install. The FM-1 restarts on DREADNOUGHT's screen.

**RECOVERY** If something goes wrong: hold OCT- and OCT+ together for 5 seconds. The FM-1 goes into update mode and the flasher can install again, or put the stock firmware back with M-VAVE's tools. docs/RECOVERY.md on GitHub has every way back. Updates keep your sounds, patterns and samples.

# 21 QUICK REFERENCE

**The combinations, on one page.**

| PRESS | FOR |
|---|---|
| HOME | the eight tracks; key 1-8 opens one |
| FX on HOME | the instrument picker |
| SEQ | this instrument's sequencer; again: back |
| SEQ + white 1-8 | that track's sequencer / instrument |
| SEQ + black 1-8 | that track's sound pages |
| HOME + SELECT | next / previous instrument |
| SEL ... GLO again | the page's second view |
| GLO ×2 / ×3 / ×4 / ×5 | FX · SETUP · MIDI · PROJECT |
| ARP ×1 / ×2 / ×3 | BRK · BRK2 · SLICE |
| LFO again (REESE, SUB) | LFO 1 ... 4 |
| FX held ½ s | the FX page; again: back |
| FX on ROLL / DRUM | STEP · KEYS · PLAY mode |
| black 10 / 11 held + white 1-4 | track 1-4 bars long |
| REC held + knob | record quantise |
| step held + knobs | NOTE LENGTH VEL COND; FX: SLIDE ACCENT RETRIG RATE; FX: RT LEN RT VEL CHORD |
| step held + another step | move it there (SAVE held: copy) |
| step held + REC + last step | velocity ramp |
| SAVE + PRESETS | store to USER 1-8 |
| SAVE + FX | name the project / sound |
| SAVE + white 1-8 | pattern 1-8 |
| SAVE + black 10 / 11 | copy / paste the pattern |
| SAVE + OCT- / OCT+ | copy / paste the bar |
| SAVE + REC | ×2: double the pattern |
| HOME held (sequencer) | FILL |

| PRESS | FOR |
| --- | --- |
| **black 9 held 1 s** | clear the track |
| **OCT-** + **OCT+** **5 s** | update mode |